Mini Project Review 1

Title: Mini Project Topic Selection and Review 1

1. Project Details

- **Project Title:** Smart Home Appliance Energy Management System: A Data-Driven Hybrid AI Approach for Climate & Load Control
- **Student Details:**

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- **Subject:** Data Science Lab
- **Academic Year:** 2026–27
- **Department:** Information Technology

2. Problem Statement

Background of the Problem

In tropical and high-humidity micro-climates, standard household cooling appliances operate in total functional isolation. High-power appliances like Inverter Air Conditioners (ACs), BLDC Ceiling Fans, and Refrigerators run on isolated internal thermostats without exchanging data or adjusting to dynamic environmental conditions.

During hot and humid monsoon weather ("moist" weather), occupants frequently over-cool rooms with ACs simply to extract moisture, leading to severe energy inefficiency and high electricity bills.

Who Faces This Problem?

- **Urban and Suburban Households:** Families in high-humidity tropical regions facing excessive monthly power bills due to uncoordinated cooling appliances.
- **Inverter/Battery-Reliant Homes:** Residents experiencing abrupt battery inverter tripping during frequent power outages caused by simultaneous appliance compressor inrush currents.
- **Energy Management Researchers & Utilities:** Smart grid planners seeking software-based residential load shedding and peak-demand optimization strategies without expensive hardware upgrades.

Why Is the Problem Important?

Individual appliances cannot optimize household energy footprint independently. Treating relative humidity strictly through temperature drops leads to excessive power usage. Coordinating thermal comfort through wind-chill effects (BLDC Fans), dehumidification modes (Inverter ACs), and load-deferral (Refrigerator defrost cycles) significantly lowers residential energy consumption while ensuring uninterrupted operation on solar/inverter backup.

Existing Difficulty

Existing home automation systems operate via rigid, manual, or hard-coded rule sets (if-else triggers) that fail to capture:

- Human comfort constraints expressed in natural, continuous language.
- Probabilistic occupancy uncertainty in physical space.
- Multi-variable trade-offs between relative humidity, temperature, and power grid limitations.

Therefore, the proposed project addresses the following core problem:

How can Natural Language Processing, Bayesian reasoning, and Fuzzy Logic control be integrated into a hybrid software framework to dynamically manage appliance energy consumption, eliminate over-cooling, and prevent battery inverter tripping during power grid outages?

3. Proposed Project Title

Smart Home Appliance Energy Management System: A Data-Driven Hybrid AI Approach for Climate & Load Control

The proposed software-driven platform processes multi-modal time-series micro-climate data and human comfort commands through a 3-stage artificial intelligence architecture:

1. **Cognitive NLP Engine:** Extracts human intent and thermal comfort boundaries from natural text prompts.
2. **Bayesian Belief Network:** Infers room occupancy probabilities $P(\text{Occupancy} \mid E)$ under spatial motion uncertainty to avoid conditioning vacant spaces.
3. **Fuzzy Logic Control Engine:** Evaluates continuous micro-climate variables (Ambient Temperature, Relative Humidity, Grid Power State) using De Morgan's fuzzy rules to dynamically adjust AC Compressor Load (%), AC Mode (Cool vs. Dry), BLDC Fan Speed (Steps 1–5), and Refrigerator Defrost Timing.

4. Research Gap / Existing System Limitation

From the literature survey [1–6], existing energy management solutions primarily focus on standalone appliance scheduling, direct temperature threshold triggers, or high-cost physical IoT sensor deployments [1, 2]. They fail to address the combined effect of high humidity and thermal load on battery inverters during grid outages [1, 5]. Standard systems treat relative humidity as a secondary variable, causing AC units to run at high compressor loads when operating in Dry Mode would achieve comfort at a fraction of the power [3, 6]. Furthermore, existing implementations lack probabilistic reasoning to handle motion sensor uncertainty and do not offer intuitive natural language interaction for end users [1, 2, 4]. The proposed system fills this gap by integrating a natural language intent parser [2, 3], Bayesian occupancy inference [1], and multi-variable Fuzzy Logic control into an integrated, interactive Streamlit simulation environment [1, 5].

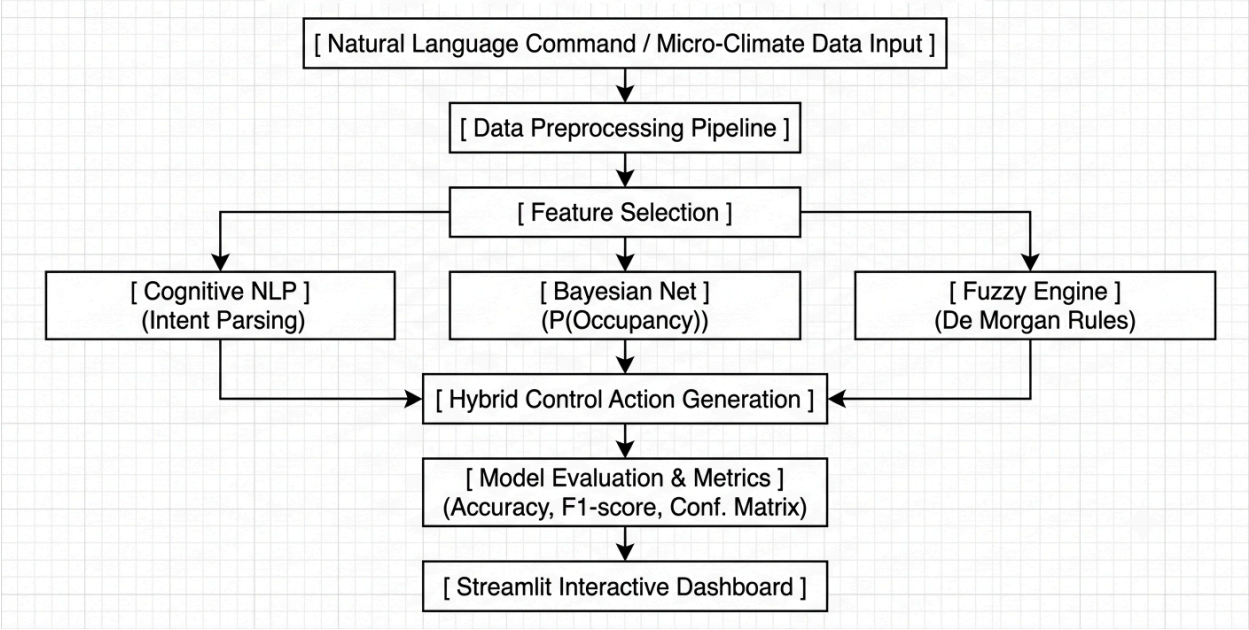
5. Objectives of the Project

1. To study multi-modal time-series micro-climate datasets and existing AI control techniques for home energy optimization.
2. To build a Cognitive NLP engine capable of parsing natural human text queries into operational thermal comfort parameters.
3. To develop a Bayesian Belief Network for inferring room occupancy probability under motion sensor uncertainty.
4. To implement a Fuzzy Logic Control Engine for multi-appliance load throttling, mode selection (Cool vs. Dry), and inverter overload prevention.
5. To evaluate system performance using standard evaluation metrics (Accuracy, Precision,

Recall, F1-score, and Confusion Matrix) and present a dashboard using Streamlit.

6. Proposed Methodology

The proposed system follows a step-by-step data processing, probabilistic reasoning, fuzzy inferencing, and visualization pipeline.



Brief Description of Each Step

- **Data Collection:** A 30-day (720-hour) micro-climate time-series dataset (smart_home_climate_data.csv) simulating tropical temperature swings ,sensor triggers, and power grid status (Mains vs. Inverter) is generated.
- **Data Preprocessing:** Missing values are handled, continuous climate variables are normalized, and textual human commands are lowercased, tokenized, and mapped to domain features.
- **Feature Selection:** Key indicators impacting appliance power state—such as ambient_temp, relative_humidity, motion_sensor, and grid_power—are isolated for inferencing.
- **Model Training & Fitting:** ML models (Random Forest, Decision Trees, Naive Bayes, KNN) and Fuzzy Rule bases are trained and fitted on the multi-modal dataset.
- **Testing & Evaluation:** System outputs are evaluated using multi-class metrics including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix visual maps.
- **Digital Footprint & Energy Dashboard:** Results, real-time slider simulations, grid failover switches, and performance plots are served through an interactive Streamlit web application.

7. Proposed Technologies / Tools

- **Python 3.10+**: Core programming language for data engineering, ML modeling, and control logic.
- **Scikit-Learn**: Machine learning classifier training, data splitting, and metrics evaluation.
- **NumPy & Pandas**: Time-series dataset generation, manipulation, and numerical processing.
- **Matplotlib & Seaborn**: Exploratory Data Analysis (EDA) plotting and confusion matrix heatmap generation.
- **Streamlit**: Interactive web dashboard development for live user simulation and parameter tuning.
- **Scikit-Fuzzy / Rule Engine**: Triangular/trapezoidal fuzzy membership function processing and defuzzification.
- **NLTK / Regex**: Natural Language Parsing of user comfort commands and intent extraction.
- **Google Colab / Jupyter**: Interactive experimentation, EDA execution, and comparative model testing.

8. Expected Outcome

The proposed system will successfully optimize household energy consumption across Inverter ACs, BLDC Fans, and Refrigerators under varying micro-climate conditions and grid states. The system expects to achieve:

1. **Zero Inverter Overload Trips:** 100% suppression of simultaneous heavy compressor starts during grid outages (grid_power = 0).
2. **Dehumidification Efficiency:** Automatic switching to AC Dry Mode during high-humidity monsoons, reducing over-cooling and saving power.
3. **High ML Classification Performance:** High prediction accuracy (95%) across multi-class operating modes verified using Confusion Matrices, Precision, Recall, and F1-Scores.
4. **Interactive GUI:** A fully functional Streamlit dashboard allowing users to input natural language commands, adjust real-time climate sliders, toggle power grid states, and visualize AI outputs.

9. References

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